PERFORMANCE

⚠ CRITICAL

DATABASE QUERY EXPLOSION

An unindexed query slipped through code review and reached production. Under normal load it was fine. Under scale it performs a full table scan on every request. The database CPU is pinned at 100% and everything is timing out.

Database performance is one of the most common scaling failure modes because queries that appear fast in development — against small datasets — can be catastrophically slow in production at scale. The N+1 query problem (executing a query per item in a loop) and missing indexes are the two most frequent culprits. Prevention requires: query analysis in CI using EXPLAIN plans against production-representative data volumes, database slow query logging with automatic alerting on queries exceeding thresholds, read replicas for reporting workloads, and connection pooling to manage database concurrency. The deeper DevOps principle is that staging environments must mirror production data volumes — a staging database with 10,000 rows cannot reveal performance problems that only emerge at 100 million rows.

Query performance does not degrade linearly. It collapses suddenly at scale — and it always hits on a Friday.

TEAM DISCUSSION

Are slow queries caught in CI? Do you run query analysis against production-scale data volumes before deploying?